

Mohit Vaid

The George Washington University | [website](#) | m.vaid@gwu.edu | [LinkedIn](#) | [Github](#)

RESEARCH INTEREST

Theoretical and applied cryptography; recent focus on Functional Encryption, Distributed Point Functions, Private Set Intersection, Post-Quantum Cryptography ; Differential Privacy

EDUCATION

The George Washington University

Washington, DC

Doctor of Philosophy, Cryptography

August 2024 - Present

- Relevant Coursework: Design and Analysis of Algorithms, Introduction to Cryptography, Differential Privacy, Advanced Software Paradigms

The Johns Hopkins University

Baltimore, MD

Master of Science

August 2023-May 2024

- Relevant Coursework: Practical Cryptography, Security and Privacy in Computing, Intrusion Detection, Intro to Algorithms

Indian Institute of Technology Kanpur

Kanpur, India

Master of Technology

August 2020-July 2022

- Relevant Coursework : Numerical Methods, Probability and Statistics

National Institute of Technology Kurukshetra

Kurukshetra, India

Bachelor of Technology

July 2012-July 2016

- Relevant Coursework: Engineering Mathematics, Computer Programming

PRESENT PROJECTS

Updatable Private Set Intersection

Oct 2025 - present

- Developing multi-party UPSI protocol with succinct communication and computation complexity

Functional Encryption

June 2025- present

- Exploring the feasibility of multi-client functional encryption for restricted function classes.
- Developing a single-input functional encryption scheme for the inner-product function class with improved computational efficiency under guidance of Prof Arkady Yerukhimovich and Prof Adam O'Neill.

Distributed Point Function

August 2024 - present

- Developing protocol for multiparty distributed point function for dishonest majority under the Supervision of Prof Arkady Yerukhimovich.

PAST PROJECTS

Real Time Proctoring Application

January 2025 - May 2025

- Developed a Python-based proctoring system compatible with any video conferencing platform leveraging MediaPipe Face Mesh and OpenCV for real-time blink rate and head movement analysis to

detect cheating behavior during online interviews, featuring adjustable detection thresholds and automated session logging. [link](#)

Blockchain Based Ransomware Proof-of-Concept

January 2024-May 2024

- Leveraged Ethereum smart contracts to build an automated escrow service for fair exchange mechanism whilst exploring technical feasibility and security implications under the supervision of Prof Mathew Green at the Johns Hopkins University. [link](#)

Padding Oracle attack on E2E Messaging Client

January 2024- May 2024

- Developed with one other an End to End Messaging Client based on ECDH, ChaCha20 and ECDSA, and conducted padding oracle attacks to study the security risks for the client, under Prof Mathew Green at Johns Hopkins University. [link](#)

WORK EXPERIENCE

Dunnhumby India Limited

Gurugram, India

Junior Developer

July 2022 -July 2023

- Developed UI and API based upon react and C# respectively for displaying the real time Jobs and also the options to cancel, delete or modify Jobs

HSCC India Limited

Noida, India

Project Management

July 2016 - July 2019

TEACHING EXPERIENCE

Design and Analysis of Algorithms, GWU Washington DC

August 2025-December2025

Numerical Methods , IIT Kanpur India

August 2021-December 2021

Atmospheric Physics and Chemistry, IIT Kanpur India

January 2022 - May 2022

CONFERENCES/WORKSHOPS ATTENDED

Secure Computation Workshop at Simon's Institute 2025

Berkeley, USA

PETS 2025

Washington DC, USA

Indocrypt 2024

Chennai, India